GinCol Lab · Aula

Student worksheet

Mathematics I · 1st year of Bachillerato · The real numbers · Class 6 of 6 · With Ximena

Properties of logarithms and change of base

Class video: https://youtu.be/pcChkHXXATw

The code leads to the video.

Name: _____ Date: _____

The video is in Spanish, with English subtitles. In Spain the decimal separator is a comma: the board writes $3{,}844 \cdot 10^{12}$ for $3.844 \cdot 10^{12}$. This sheet uses the decimal point.

Before watching the video

1. A sheet folded in half once has 2 layers. How many layers does it have after 3 folds? And after 10?

2. Write 2^3 as a product of twos. How many layers does the sheet have after 3 folds, and what does that have to do with 2^3?

3. Revision of the previous class: what is $\log_2 8$? With the product rule, what is $\log_2 (8 \cdot 4)$?

While you watch the video

1. What is the logarithm of a product equal to? Check it with $\log_2 (4 \cdot 8)$.

2. **Your bet.** How many times do you think you have to fold a 0.1 mm sheet for it to reach the Moon? Write it here: _____

3. Why, after n folds, does the sheet have 2^n layers? How many layers are needed to reach the Moon?

4. In $2^n = 3.844 \cdot 10^{12}$, where is the unknown? Why can't you work it out in your head?

5. With the product rule, why is $\log (2^3) = 3 \cdot \log 2$? Why, on the slide rule, is 8 at 0.9 if 2 is at 0.3?

6. **The power rule.** Write it in words and in symbols. What happens to the exponent?

7. What is the difference between the logarithm of a power, $\log_2 (4^3)$, and the power of a logarithm, $(\log_2 4)^3$? What is each one?

8. **Your answer.** Why can you take logarithms of both sides of an equation? Take logarithms of both sides of $2^n = 3.844 \cdot 10^{12}$, bring n down and solve for it.

9. What is n? Why are 42 folds needed and not 41? How did your bet do?

10. Why does the last fold "double it all"?

11. Can a sheet really be folded 42 times? What did Britney Gallivan achieve in 2002?

12. **The change of base.** Write it in words and in symbols. Which base goes on top and below? Why can't the base be 1?

13. Work out $\log_2 3$ with the change of base. Does it match what you saw in the previous class?

14. Exercise 3 from the previous class: between which two whole numbers is $\log_2 10$, and why? What is it with the change of base?

15. Why is log 3 / log 2 not the same as log (3 / 2)?

16. **Your answer.** Between which two whole numbers is $\log_3 20$? What is it with the calculator?

17. What is the half-life of carbon-14? How long does it take for a tenth to be left, and why does $\log_2 10$ come up again?

The power rule, with gaps

Work out $\log_2 (8^2)$, filling in the gaps without looking at the video.

1 · The rule. The logarithm of a power is the _____ times the logarithm of the number being raised: $\log_2 (8^2) =$ ___ · $\log_2 8$.

2 · The logarithm of 8. What number must you raise 2 to in order to get 8? $\log_2 8 =$ ___.

3 · The result. $\log_2 (8^2) =$ ___ · ___ = ___. Check it: $8^2 = 64$, and 2^__ = 64.

Now, without help:

1. $\log (10^4)$

2. $\log_3 (9^5)$

The change of base, with gaps

Work out $\log_5 30$, filling in the gaps without looking at the video.

1 · First, the interval. 5^2 = ___ and 5^3 = ___, so $\log_5 30$ is between ___ and ___.

2 · The rule. $\log_5 30$ = log ___ / log ___.

3 · With the calculator (the log key opens a bracket: close it after each number). log 30 ≈ _____ and log 5 ≈ _____, so $\log_5 30$ ≈ _____. Does it fall in the interval from step 1?

Practice

1. Without a calculator: 2 · log 5 + log 4.

2. With the calculator: $\log_2 1000$. First, see which two whole numbers it lies between.

3. A bacterium divides in two every 20 minutes. Starting with just one, how long does it take until there are more than a million?

4. **One more, with the sheet.** In the video, the sheet passes 100 km with 30 folds. Check it with logarithms: how many 0.1 mm layers are needed for 100 km?

5. **One more, with the mistake.** A classmate writes $\log (2^5) = (\log 2)^5$. Work out both sides with the calculator and explain to them, with the power rule, where the mistake is.

6. **A challenge, without a calculator.** With $2^{10} \approx 10^3$, work out in your head roughly how thick a 0.1 mm sheet folded 42 times would be. Hint: $2^{42} = 4 \cdot (2^{10})^4$.

Keep thinking

Why does our intuition fool us so badly with things that grow by multiplying?

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